Tonight's Study Plan

MYP 3 Mathematics - Unit 1: Numbers and Perspectives

Topics: **Real number systems & perfect squares**, **Exponents**, **Scientific notation**.

Time	What to do	Done
30 min	**Part 1 - Exponents:** learn the 5 laws, then solve the practice sheet (Q1-Q10)	[ ]
10 min	Break	[ ]
25 min	**Part 2 - Scientific notation**	[ ]
25 min	**Part 3 - Real number systems & perfect squares**	[ ]
10 min	Break	[ ]
30 min	**Part 4 - Quick quiz** (no peeking!), then check with the answer key	[ ]
10 min	**Part 5 - Read the 'Common mistakes' list**, then sleep early	[ ]

PART 1 - EXPONENTS (most important)

The 5 laws

If you see...	Do this	Example
Multiply, same letter	**ADD** the exponents	$a^2 \times a^3 = a^5$
Divide, same letter	**SUBTRACT** (top - bottom)	$a^7 / a^2 = a^5$
Bracket with a power	**MULTIPLY** the exponents	$(a^2)^3 = a^6$
Negative exponent	**MOVE** it to the other floor, make it positive	$a^{-2} = 1 / a^2$
Zero exponent	Answer is **1**	$7^0 = 1$

5 steps for any big question

- **1. Open the brackets:** the power goes to EVERYTHING inside, numbers too: $(2x^3)^4 = 2^4x^{12} =$ **$16x^{12}$**
- **2. Dividing two fractions?** Keep - Change - Flip: keep the first, change / to x, flip the second.
- **3. Numbers alone:** multiply or divide the ordinary numbers.
- **4. Each letter alone:** add its exponents on top, then subtract its exponents on the bottom.
- **5. Clean up:** any negative exponent moves to the other floor; a letter with exponent 0 disappears.

4 traps to remember:
- $(2a^3)^2 =$ **$4a^6$**, NOT $2a^6$
- $a^6 / a^{-2} = a^{6-(-2)} =$ **a^8** (minus a minus = plus)
- $(6/7)^{-2} = (7/6)^2 =$ **49/36** (negative exponent flips the fraction)
- $-9^0 =$ **-1** but $(-9)^0 =$ **1**

PART 2 - SCIENTIFIC NOTATION

Form: a number from 1 up to (but not including) 10, times 10 to a power: **a x 10^n**, where 1 <= a < 10.

Correct: 3.5 x 10^4 Wrong: 35 x 10^3 (35 is bigger than 10)

Ordinary number -> scientific notation

- **1.** Move the decimal point until only **one digit** (not zero) is before it.
- **2.** Count how many places you moved. That is the exponent.
- **3. Big** number -> exponent **positive**. **Small** number (0.00...) -> exponent **negative**.
- Examples: 45,000 = **4.5 x 10^4** 0.0072 = **7.2 x 10^{-3}**

Scientific notation -> ordinary number

- Positive exponent -> move the point **right**: 3.1 x 10^5 = **310,000**
- Negative exponent -> move the point **left**: 6 x 10^{-4} = **0.0006**

Multiplying and dividing

- **Multiply:** multiply the numbers, ADD the exponents: (2 x 10^3)(4 x 10^5) = **8 x 10^8**
- **Divide:** divide the numbers, SUBTRACT the exponents: (9 x 10^7) / (3 x 10^2) = **3 x 10^5**

Fix it if the number is 10 or more: (5 x 10^3)(4 x 10^2) = 20 x 10^5 = **2 x 10^6** (the point moved 1 place left, so add 1 to the exponent).

PART 3 - REAL NUMBER SYSTEMS & PERFECT SQUARES

The number sets fit inside each other like Russian dolls:

Set	Examples
Natural	1, 2, 3, ...
Whole	0, 1, 2, 3, ... (natural numbers + zero)
Integers	..., -2, -1, 0, 1, 2, ... (negatives too)
Rational	Can be written as a **fraction**: 1/2, -3, 0.75, 0.333...
Irrational	**Cannot** be written as a fraction: pi, √2, √5
Real	All of the above together

Rational or irrational?

- Decimal that **stops** (0.25) or **repeats** (0.333...) -> **rational**.
- Decimal that **never stops and never repeats** (pi = 3.14159...) -> **irrational**.
- √ of a perfect square (√25 = 5) -> **rational**. √ of a non-perfect square (√7) -> **irrational**.

Perfect squares (learn by heart)

1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225

Estimating a square root

Find the two perfect squares it sits between: √30 is between √25 = 5 and √36 = 6, so **√30 is about 5.5**.

PART 5 - COMMON MISTAKES (read before bed and in the morning)

- The number inside the bracket gets the power too: $(2a)^2 = 4a^2$.
- Minus a minus is a PLUS: 6 - (-2) = 8.
- A negative exponent on a fraction FLIPS the fraction.
- Final answers must have NO negative exponents.
- In scientific notation the front number must be from 1 up to (but not including) 10.
- Write EVERY step in the exam - steps earn marks even if the final answer is wrong.

Quick Quiz

Exponents - Scientific Notation - Real Numbers

Name: _____ Score: _____ / 14

Show your working. Write final answers with positive exponents only. **12 or more correct = ready for the test!**

1. Simplify: $x^5 \times x^{-2}$

2. Simplify: $(3a^2)^3$

3. Find the value (as a fraction): 2^{-4}

4. Find the value (as a fraction): $\left(\frac{2}{5}\right)^{-2}$

5. Simplify (positive exponents only): $\frac{12x^5y^{-2}}{4x^2y^3}$

6. Find the value: $-2^0 - (-2)^0$

7. Write in scientific notation: 0.00058

8. Write as an ordinary number: 4.07×10^5

9. Calculate. Give the answer in scientific notation: $(6 \times 10^4)(5 \times 10^3)$

10. Calculate. Give the answer in scientific notation: $\frac{8 \times 10^9}{2 \times 10^4}$

11. Is each number rational or irrational? $\sqrt{49}$ $\sqrt{10}$

12. Between which two whole numbers is $\sqrt{50}$?

13. Is -4 a whole number? Name all the sets it belongs to.

14. List all the perfect squares between 50 and 150.